5.8 Stokes' Theorem Worksheet

1. Suppose $\mathbf{F} = \left\langle e^{xy}, e^{xz}, zx^2 \right\rangle$ and σ is the half of the ellipsoid $4x^2 + y^2 + 4z^2 = 4$ that lies to the right of the *xz*-plane, oriented in the direction of the positive *y*-axis. Use Stokes' Theorem to evaluate $\iint\limits_{\sigma} \operatorname{curl}(\mathbf{F}) \cdot d\mathbf{S}$.

2. Suppose $\mathbf{F} = \left\langle 1, x + yz, xy - \sqrt{z} \right\rangle$ and C is the boundary of the part of the plane $3x + 2y + z = 1$ in the first octant. Given that C is oriented counterclockwise as viewed from above, use Stokes' Theorem to evaluate $\int\limits_{C} \mathbf{F} \cdot d\mathbf{S}$.

3. Let $\mathbf{F} = \sqrt{1+x}\mathbf{i} + \sinh^{-1}(y)\mathbf{j} + \ln\left(\sqrt{z^2+1}\right)\mathbf{k}$ and C be the intersection of the cone $z = \sqrt{x^2+y^2}$ and the sphere $x^2+y^2+z^2=2$ with a counterclockwise orientation as you look down the positive z-axis. Evaluate $\int_C \mathbf{F}\cdot d\mathbf{r}$.

4. Let $\mathbf{F}(x,y,z) = -3y^2\mathbf{i} + 4z\mathbf{j} + 6x\mathbf{k}$ and C be the triangle in the plane $z = \frac{1}{2}y$ with vertices $(2,0,0)$, $(0,2,1)$, and $(0,0,0)$ with a clockwise orientation as you look down the positive z-axis. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$.

5. Consider the vector field $\mathbf{F}(x,y,z)=\langle yz,-xz,yx\rangle$. Let σ be the portion of the sphere $x^2+y^2+z^2=13$ where $x\le 3$.

(a) Find the flux of $\operatorname{curl}(\mathbf{F})$ through σ with respect to outward pointing normals.

(b) How would your answer from part (a) change if we instead worked with the portion of the sphere $x^2+y^2+z^2=13$ where $x\ge 3$.

6. Let $\mathbf{F}(x,y,z)=(z+\sin(x))\mathbf{i}+(x+y^2)\mathbf{j}+(y+e^z)\mathbf{k}$ and C be the intersection of the paraboloid $z=8-x^2-y^2$ and the paraboloid $z=3x^2+y^2$ with counterclockwise orientation as you look down the positive z-axis. Evaluate $\int\limits_C \mathbf{F}\cdot d\mathbf{r}$.

7. Let σ be the surface parametrized by $\mathbf{r}(u,v)=\langle 2-2v^2, v\cos(u), v\sin(u)\rangle$ for $0\le u\le 2\pi$ & $0\le v\le 1$.

a) For the vector field $\mathbf{F}=\langle 0,-z,y\rangle$, directly evaluate $\iint_\sigma \operatorname{curl}(\mathbf{F})\,d\mathbf{S}$, where **n** is a unit normal that points in the positive x-direction.

b) Check your answer to part (a) by computing $\iint_{\sigma} \operatorname{curl}(\mathbf{F})\, d\mathbf{S}$ using Stokes' Theorem.